

X-LowSculptor - Plugin Design Spec

Version 1.0 - February 16, 2026

Overview

X-LowSculptor is a bass-focused processor inspired by a Michael Brauer style split workflow. It splits the signal into two bands (BODY and NECK), provides dedicated processing per band, and recombines the result with professional mix controls (gain staging, per-band mute/bypass, and output metering).

Signal Flow (Start to Finish)

```
IN
-> IN TRIM (fader)
-> INPUT METER + SWEET SPOT (green range)
-> CROSSOVER (Freq variable, default 150 Hz)
    |-- BODY (low band)
    |   -> BODY GAIN (fader)
    |   -> MONO LOWS (button, optional)
    |   -> BODY BYPASS? (button)
    |       -> if OFF:
    |           -> SQUEEZE (RS124-inspired comp) [parallel internal]
    |           -> OOMPH (subharmonics generator) [parallel internal]
    |           -> PUMP (ducking) [serial]
    |               - Mode: SYNC (LFO tempo) / SC (real sidechain)
    |               - Amount (knob)
    |       -> if ON: pass BODY clean
    |   -> BODY MUTE? (button)
    |-- NECK (high band)
    |   -> NECK GAIN (fader)
    |   -> NECK BYPASS? (button)
    |       -> if OFF:
    |           -> CLAMP (FET compression) [serial]
    |           -> GRIT (saturation) [serial]
    |       -> if ON: pass NECK clean
    |   -> NECK MUTE? (button)
-> SUM (BODY + NECK)
-> OUT GAIN (fader)
-> OUTPUT METER
-> OUT
```

Routing Rules

- **Mute wins over Bypass:** if a band is muted, it outputs silence regardless of bypass state.
- **Bypass per band:** bypass skips that band's DSP but preserves the band audio.
- **Mono Lows:** when enabled, BODY is summed to mono before its processing and recombination.
- **PUMP SC fallback:** if PUMP is set to SC but sidechain is absent or silent, fallback to SYNC (recommended).

Controls and Options

Global

Control	Type	Notes / Default
IN TRIM	Fader	Gain staging into the processor; drives subsequent sections.
Input Meter + Sweet Spot	Meter	Indicates correct input range (green).
Crossover Freq	Knob	Variable split frequency; default 150 Hz.
OUT GAIN	Fader	Final output trim.
Output Meter	Meter	Shows final level.

BODY (Low band)

Control	Type	Notes
Body Gain	Fader	Level control for low band before processing.
Body Bypass	Button	Skips Squeeze/Oomph/Pump for BODY.
Body Mute	Button	Silences BODY output.
Mono Lows	Button	Sums BODY to mono for solid low-end.
SQUEEZE	Knob	One-knob RS124-inspired compression (parallel internal).
OOMPH	Knob	Subharmonics generator amount (parallel internal).
PUMP Mode	Switch	SYNC (tempo LFO) or SC (real sidechain).
PUMP Amount	Knob	Depth of ducking / pumping.

NECK (High band)

Control	Type	Notes
Neck Gain	Fader	Level control for high band before processing.
Neck Bypass	Button	Skips Clamp/Grit for NECK.
Neck Mute	Button	Silences NECK output.
CLAMP	Knob	One-knob FET compression (serial).
GRIT	Knob	Saturation for definition (serial).

Design Notes

- **Per-band MUTE/BYPASS** are required for fast auditioning (soloing BODY or NECK).
- **SQUEEZE and OOMPH are parallel internally** to maintain musical low-end stability and simplify UI.
- **PUMP supports two workflows**: SYNC for electronic/synth bass and SC for kick-driven mixing.
- **Channel formats**: the plugin supports mono and stereo instantiation ({1,1} and {2,2}).
- **Status**: this document captures the agreed functional specification for X-LowSculptor and will drive DSP block implementation.